In [4]:
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# Train a supervised ML model to classify malware vs benign files
# Dataset file required: malware_dataset.csv

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# 1. Load dataset
df = pd.read_csv("malware_dataset.csv")

print("First 5 rows of dataset:")
print(df.head())

# 2. Separate input features and target label
X = df.drop(["file_id", "label"], axis=1)
y = df["label"]

# 3. Encode labels: benign/malware into numbers
encoder = LabelEncoder()
y_encoded = encoder.fit_transform(y)

# benign = 0, malware = 1, depending on alphabetical order

# 4. Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(X,y_encoded,test_size=0.30,random

# 5. Normalize numeric features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 6. Train supervised ML model
model = RandomForestClassifier(n_estimators=100,random_state=42)
model.fit(X_train_scaled, y_train)

# 7. Predict test data
y_pred = model.predict(X_test_scaled)

# 8. Evaluate model
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test,y_pred,target_names=encoder.classes_))

# 9. Test with a new file sample

new_file = [[1400, 170, 8, 7.9, 0, 1, 23]]
# Use the same feature names as training data
new_file = pd.DataFrame(new_file, columns=X.columns)
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new_file_scaled = scaler.transform(new_file)
prediction = model.predict(new_file_scaled)
print("\nNew file prediction:", encoder.inverse_transform(prediction)[0])
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First 5 rows of dataset:
  file_id  file_size_kb  num_imports  num_sections  entropy  has_signature  \
0    F001           120           45             5      5.1              1
1    F002           980          120             8      7.8              0
2    F003           300           65             6      5.6              1
3    F004          1500          180             9      8.2              0
4    F005           220           40             4      4.9              1

   has_packer  suspicious_api_count    label
0           0                     2   benign
1           1                    18  malware
2           0                     4   benign
3           1                    25  malware
4           0                     1   benign

Accuracy: 1.0

Confusion Matrix:
[[3 0]
 [0 3]]

Classification Report:
              precision    recall  f1-score   support

      benign       1.00      1.00      1.00         3
     malware       1.00      1.00      1.00         3

    accuracy                           1.00         6
   macro avg       1.00      1.00      1.00         6
weighted avg       1.00      1.00      1.00         6


New file prediction: malware
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In [ ]: